

NITIN TEWARI

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Summary

AI & Machine Learning Developer with 2+ years at **PwC** building scalable **ML** models, **ETL** pipelines, and LLM-powered **Generative/Agentic AI** solutions. Proficient in Python, TensorFlow, LangChain, RAG, NLP, and prompt engineering, with a strong foundation in **data wrangling** and analytics to drive actionable business insights .

Education

Indiana University Bloomington

Aug 2024 – May 2026

Master of Science in Data Science

Bloomington, USA

- Coursework: Machine Learning, Statistics, Data Mining, NLP

SRM Institute of Science and Technology, Chennai

July 2018 – May 2022

Bachelors of Technology in Computer Science

Chennai, India

- Coursework: Statistical Analysis, Data Structures and Algorithms, Database Systems

Experience

Indiana University Bloomington

Jan 2026 – May 2026

AI Research Assistant

USA

- Built a **4-agent** autonomous pipeline (Search → Reader → Writer → Critic) using **LangChain LCEL** to retrieve live **web data**, generate structured research reports, and **auto-score** output quality in a single query
- Extended the pipeline with automated PDF report generation using **FPDF**, enabling one-click downloadable research

PwC

Sep 2022 – Jan 2024

Data Scientist

Bangalore, India

- Engineered predictive models using **Logistic Regression** and **Random Forest** in Python and R, achieving 85% accuracy; integrated with **Git** and **CI/CD** pipelines for automated testing and version control
- Orchestrated a hybrid recommendation engine using **Apache Spark, ALS, and collaborative filtering**, contributing to a **25% increase in sales**, validated through **A/B testing**
- Deployed **NLP** sentiment pipeline fine-tuning **DistilBERT** on customer reviews using Hugging Face and AWS SageMaker, achieving **93% F1** -a 5-point improvement over **TF-IDF** baseline
- Designed and maintained **Power BI dashboards and visualizations** integrating data from **AWS S3, SAP, and SQL systems**, enabling executive reporting and improving forecasting accuracy by **15%**
- Collaborated with **business stakeholders**, engineers, and analytics teams to translate requirements into scalable analytical solutions and executive-ready reports
- Streamlined **SQL** query development, documentation, and data transformation logic using AI-assisted coding tools (**GitHub Copilot**)

PwC

March 2022 – Aug 2022

Associate Intern

Bangalore, India

- Performed **exploratory data analysis, data cleansing, and statistical analysis**, in Python and SQL across large, complex datasets, improving data accuracy by **15%**
- Managed **documentation of datasets, reports, and data definitions** to support data governance

Projects

Healthcare MediBot – RAG | LLM, NLP, LangChain, FAISS |

Dec 2025

- Implemented a **RAG**-powered application using **FAISS** and **LangChain** to extract entities from **electronic health records (EHRs)**, improving data retrieval speed by **40%** and reducing manual processing
- Applied **NLP** techniques with spaCy and NLTK to preprocess and analyze clinical text data, enabling accurate symptom recognition from unstructured health records

Real-Time Data Engineering Pipeline | Kafka, AWS, Python, SQL |

May 2025

- Architected an **ELT pipeline** streaming **10K+** events/hour via Apache Kafka on AWS EC2; automated **schema discovery** and **data lineage** using AWS Glue, improving query efficiency by **30%**

Technical Skills

Languages: Python, SQL, R, Java

Data: ETL, Data Modeling, Data Validation, Data Quality, Data Cleansing, Ad-hoc Reporting, risk management

Visualization: Power BI, Tableau, Excel (Pivot Tables, vLookups, Formulas)

Cloud & AI/ML: GCP, AWS (S3, EC2, Glue), Amazon Redshift, Snowflake, Databrick TensorFlow, Scikit-learn,

Tools: Alteryx, SAP, Oracle, PowerPoint, MATLAB